

鲁天宇 Andy · 活动与荣誉列表

COMMON APP

Activities（10 条，按优先级）

- #1 无人机-地面车协同调度研究（含 TAMS 暑期科研）

TYPE Research

POSITION Student Researcher [18/50]

ORGANIZATION Univ. of North Texas, Prof. Qing Yang's lab (via TAMS) [54/100]

GRADE 11, 12 | 15 hr/wk · 44 wk/yr

DESCRIPTION Integrated attention into PPO for 3-UGV Open-RMF dispatch; achieved lowest completion time & queue variance in tests; submitted 1st-author paper.~146/150]
- #2 竞速无人机设计、搭建与首飞（吉尼斯世界纪录-瞬时速度）-G12 Continue

TYPE Robotics

POSITION Hardware & Flight Lead [22/50]

ORGANIZATION Ek-introduce[32/100]

GRADE 11, 12 | 25hr/wk · 10wk/yr

DESCRIPTION Built a racing drone w/ 3 peers; owned hardware integration & flight tuning; ran the pre-flight risk checks, then performed the maiden flight [~141/150]
- #3 音频创作八年：朗诵频道 → 工程播客

TYPE Other Club/Activity

POSITION Founder, Host & Producer [24/50]

ORGANIZATION Ximalaya recitation channel (2018-Present) & 2 engineering podcasts [67/100]

GRADE 9, 10, 11, 12 | 2 hr/wk · 30 wk/yr

DESCRIPTION Posted recitation & English dubbing since 2018; launched new 23 episodes rebuilding engineering accidents & self-driving notes; received 16.8k plays [~148/150]
- #4 Jasper 高中物理社（创办）

TYPE Science/Math

POSITION Founder & President [19/50]

ORGANIZATION Jasper High School Physics Club [31/100]

GRADE 10 | 2 hr/wk · 16 wk/yr

DESCRIPTION Founded school's 1st physics club, recruiting 60+ members; ran wkly workshops on college-level topics & demos; coached Physics Bowl team to 2nd in TX [~149/150]
- #5 机翼扭转气动特性研究

TYPE Research

POSITION Researcher [22/50]

ORGANIZATION Jasper School Science Fair- A study on wing washout aerodynamics [65/100]

GRADE 10 | 10 hr/wk · 8 wk/yr

DESCRIPTION Built CFD models of twisted Boeing 737 wings (0°/3°/5° washout) in ANSYS Fluent; tested 21 cases over 0-18° attack; 5° twist eliminated wingtip stall [-143/150]

#6 STEM Bridge 风机状态监测 (UT Dallas)

TYPE Research

POSITION Team Researcher [15/50]

ORGANIZATION STEM Bridge Summer Camp, UT Dallas (Texas Assoc. of Science & Technology) [73/100]

GRADE 10 | 15 hr/wk · 4 wk/yr

DESCRIPTION Prototyped a low-cost wind turbine monitor targeting generator faults, 17% of turbine failures & 38% of operating cost, won best science education award [-152/150]

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#7 AYLUS 社区服务

TYPE Community Service (Volunteer)

POSITION Fundraising Volunteer [21/50]

ORGANIZATION Alliance of Youth Leaders in the United States (AYLUS) [54/100]

GRADE 9, 10 | 1 hr/wk · 16 wk/yr

DESCRIPTION Ran park concession stands raising \$300 for children w/ disabilities; packed book donations for Africa & meals at Feed My Starving Children [-141/150]

#8 Stephens 小学数学志愿辅导 + MathFun

TYPE Community Service (Volunteer)

POSITION Volunteer Math Tutor [20/50]

ORGANIZATION Stephens Elementary School [26/100]

GRADE 11, 12 | 1.5 hr/wk · 10 wk/yr

DESCRIPTION Tutored 20 4th-5th graders in math wkly; organized MathFun, a game-based session that pulled them into problem solving [-117/150]

#9 Research Assistant

TYPE Science/Math

POSITION Research Assistant [34/50]

ORGANIZATION University of North Texas — Autonomous-Driving Perception guided by PhD [25/100]

GRADE 11-12 | 5 hr/wk · 44 wk/yr

DESCRIPTION Built MMDetection +Transformer pipeline for 6-camera occupancy perception; localized nearby vehicles vs ego car; benchmarked 6-view vs single-view [-145/150]

#10 篮球与长期力量训练

TYPE Athletics: Club

POSITION Point Guard & Strength Training [31/50]

ORGANIZATION TAMS Basketball Club; independent strength training [51/100]

GRADE 9-12 | 5 hr/wk · 40 wk/yr

DESCRIPTION Played point guard in the UNT Open Basketball Championship; kept a near-daily lifting & running routine since middle school [-123/150]

Honors (5 条)

#1 AMC 10 Distinguished Honor Roll (top 1%); AMC 12 Distinction (top 5%); AIME 12, USAMO Qualifier

LEVEL National | Grade: 10, 11 [93/100]

#2 USA Physics Olympiad Qualifier via the F=ma Exam

LEVEL National | Grade: 10 [34/100]

#3 Physics Bowl, 2nd-Place Team in Texas, Region 11, Division 1

LEVEL State/Regional | Grade: 10 [49/100]

#4 Plano ISD Superintendent's Scholar Award

LEVEL State/Regional | Grade: 10 [56/100]

#5 Best Science Education Award, TAST STEM Bridge Summer Research Camp

LEVEL School | Grade: 10 [56/100]

ADDITIONAL INFORMATION

Activity 1:

Under Professor Qing Yang's supervision at UNT, I developed an Attn-PPO dispatcher for three UGVs in an Open-RMF airport-terminal simulation. The UAV supplied simulated target notifications but was not controlled by the policy. I integrated self-attention with PPO, ran comparative experiments, analyzed performance trade-offs, and presented weekly progress. Attn-PPO achieved the lowest completion time and queue variance in the tested single-map configuration, while greedy scheduling performed better on waiting time and movement distance. I am first author of the resulting paper, submitted to IEEE AIoT on August 1, 2026 and currently under review. Full essay:

Activity 2:

Racing drone, four-person team — I took hardware integration and flight tuning, and flew the maiden flight. TBD: 照片/视频链接

Activity 3:

"Engineering Insights" and "Self-Driving Beginner's Journal", written and recorded by me. The first takes a documented accident apart to find the engineering decision underneath it; the second logs what I am learning about autonomous driving, including what I get wrong. TBD: 链接:]

Activity 5:

This project was a two-person collaboration; I led the CFD setup, including geometry repair, meshing, and SST k-omega solver configuration in ANSYS Fluent, and co-wrote the report. Full report:

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